The Bologna Semester Plan (40 Marks Total)

- **Total Marks:** 100 Marks (100%)
- **Weight Distribution:** 40% Continuous Assessment (Classwork) — 60% Final Examination

Assessment Framework (Total: 40 Marks Continuous Assessment)

To fulfill the Bologna Process criteria for student-centered learning and continuous learning monitoring, the 40 marks allocated to classwork are distributed across diverse assessment types:

1. Mid-term Examination (10 Marks): A formal, written theoretical exam on week 7.
2. **Quizzes (10 Marks Total):** Two short tests (5 marks each) designed to maintain consistent study habits and monitor progressive understanding.
3. **Case Study Research Report (10 Marks):** An independent academic paper requiring analytical evaluation of real-world wireless architectures or protocol standards.
4. **Case Study Presentation / Seminar (5 Marks):** An oral presentation to evaluate the student's communication, professional synthesis, and peer defense skills.
5. **Attendance & Active Engagement (5 Marks):** A continuous evaluation metric assessing regular presence and qualitative contribution to weekly class technical debates and problem-solving sessions.

Case Study & Seminar Master Guideline.

1. **The Recommended Topics for Case Study & Seminar Master Guideline**

   * Students must select one of the following topics by Week 8.
   A maximum of three students/groups may select the same topic to ensure diversity during the Week 15 seminars.

   - **Architectural Evolution & Edge Computing**
     + **Topic 1:** The Transition Matrix: 5G Non-Standalone (NSA) vs. Standalone (SA) Architectures.
     + **Focus:** Analyzing the theoretical deployment differences, signaling overhead, and latency improvements when moving away from a 4G LTE core network dependency.
     + **Topic 2:** Low-Latency Execution: Cloud-Native Multi-access Edge Computing (MEC).
     + **Focus:** Evaluating how moving compute infrastructure to the base station impacts data delivery metrics in autonomous vehicular safety loops or factory automation.
   - **Next-Generation Transport & Physical Efficiencies**
     + **Topic 3:** Overcoming Wireless Fading: QUIC (HTTP/3) vs. Traditional TCP.

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    + **Focus:** Investigating how connection migration and zero-RTT handshakes mitigate the issues of physical packet drops caused by fading and user mobility.
    + **Topic 4:** Multi-Gigabit Local Overlap: Wi-Fi 7 (IEEE 802.11be) vs. 5G/6G Cellular.
    + **Focus:** A comparative analysis of channel widths, interference mitigation, and coexistence patterns in dense corporate environments.
  - **Dynamic Topologies & Sensor Deployments**
    + **Topic 5:** Infrastructure-less Logistics: Dynamic Routing in Mobile Ad-hoc Networks (MANETs).
    + **Focus:** Examining how reactive vs. proactive routing protocols handle unpredictable link breaks caused by moving nodes without fixed base stations.

2. **Technical Submission Specifications (The Report — 10 Marks)** Every student must submit a formal technical report via the university portal by the end of Week 14. The report must strictly follow these structural criteria:
   - **Format:** Standard PDF document, font size 12, single-spaced, between 4 to 6 pages.
   - **Structure:**
     i. **Introduction & Context:** Defining the selected wireless environment/problem.
     ii. **Theoretical Analysis:** Explaining the protocol mechanics or structural frameworks involved.
     iii. **Trade-off Evaluation:** Critically analyzing the benefits, drawbacks, and limitations (e.g., latency vs. energy consumption).
     iv. **Conclusion:** A concise future outlook based on your readings.
     v. References: Minimum of 3 credible sources (academic papers or industry standards).

3. **Seminar Presentation Rules (The Presentation — 5 Marks)** During Week 15, presentations will follow a strict academic format to cultivate technical defense and presentation skills:
   - **Duration:** Maximum 7 minutes for presentation + 3 minutes for peer and instructor Q&A.
   - **Slide Limit:** Recommended 5–7 visual, text-light slides.
   - **Structure:**

   Slide 1: Title & Objective.
   Slide 2: Background / Current Challenges.
   Slide 3: Core Technology / Architectural Solution.

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Slide 4: Comparative Matrix / Engineering Trade-offs.

Slide 5: Summary & Key Takeaway.

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